

[Aim: 100/100 in Maths]

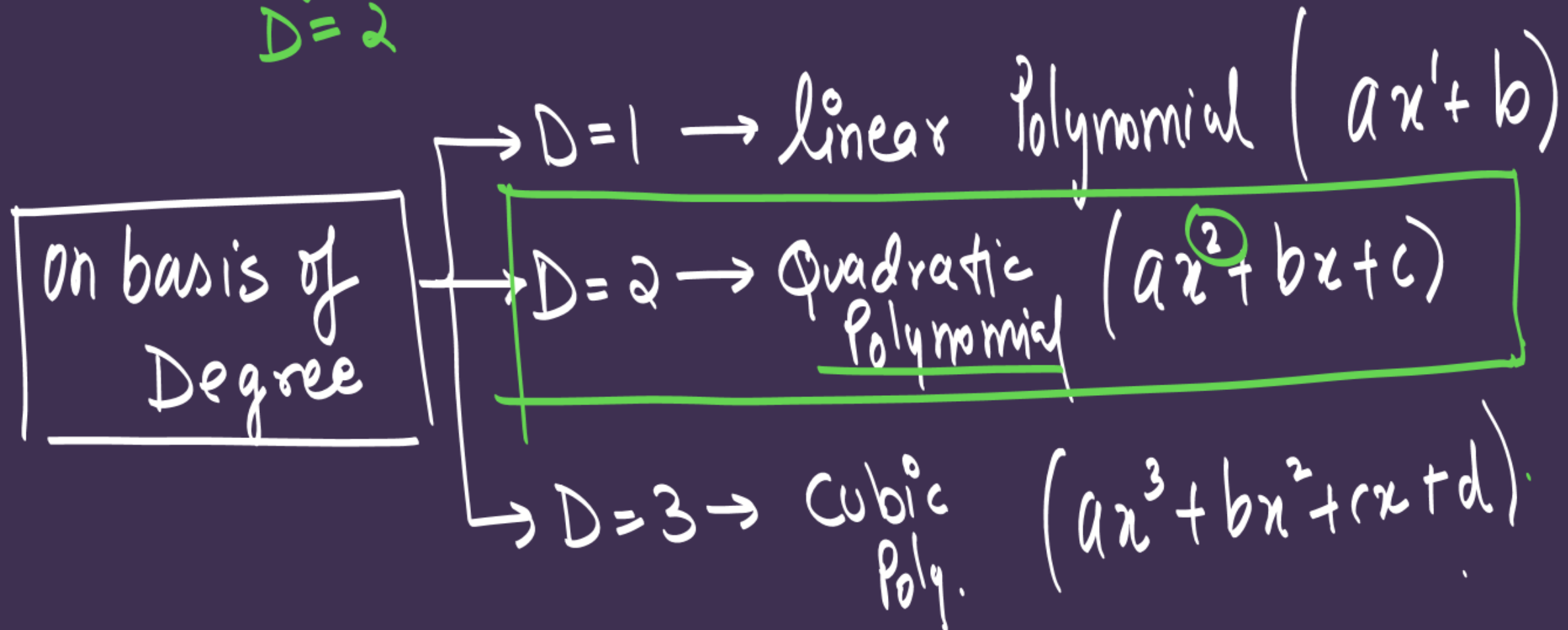
आरंभ

QUADRATIC EQUATIONS

Lecture #1

Quadratic Equations

$D=2$



$(=)$

$$ax^2+bx+c=0$$

Quadratic Equation

पहचानें ? $\Rightarrow D=2$

#LP: Which of the following are quadratic equations?

(i) $x^2 + 6x - 4 = 0$

$D=2$ ✓
 $Q.E$ ✓

(ii) $\sqrt{3x^2} - 2x + 1/2 = 0$

$D=2$
 $Q.E$ ✓

(iii) $(x+1)^2 = 2(x-3)$

→ further solvable

~~$D=1$ $Q.E$ ✗~~
 $x^2 + 1 + 2x = 2x - 6$

$x^2 + 1 + 6 = 0$

$x^2 + 7 = 0$

$D=2$
 $Q.E$ ✓

(iv) $x^3 - 4x^2 - x + 1 = (x-2)^3$

$x^3 - 4x^2 - x + 1 = x^3 - (2)^3 - 3(x)(2)(x-2)$

$\Rightarrow -4x^2 - x + 1 = -8 - 6x(x-2)$

$\Rightarrow -4x^2 - x + 1 = -8 - 6x^2 + 12x$

$\Rightarrow 6x^2 - 4x^2 - 12x - x + 8 + 1 = 0$

$\Rightarrow 2x^2 - 13x + 9 = 0$

$D=2$
 $Q.E$ ✓

$$(v) (x-2)(x+1) = (x-1)(x+3)$$

$$\cancel{x^2} + x - 2x - 2 = \cancel{x^2} + 3x - x - 3$$

$$\Rightarrow \underline{-x - 2} = \underline{2x - 3}$$

$$\Rightarrow -x - 2x + 3 - 2 = 0$$

$$\Rightarrow \boxed{-3x + 1 = 0}$$

$$D = 1$$

$$0 \in \mathbb{R}$$

$$(vi) (x + 1/x)^2 = 3(x + 1/x) + 4$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = 3\left(x + \frac{1}{x}\right) + 4$$

$$\Rightarrow (x)^2 + \left(\frac{1}{x}\right)^2 + 2(x)\left(\frac{1}{x}\right) = 3x + \frac{3}{x} + 4$$

$$\Rightarrow \underline{x^2} + \underline{\frac{1}{x^2}} + \underline{2} = \underline{3x} + \underline{\frac{3}{x}} + \underline{4}$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 3x - \frac{3}{x} + 2 - 4 = 0$$

$$\Rightarrow \underline{x^2} + \underline{\frac{1}{x^2}} - \underline{3x} - \underline{\frac{3}{x}} - \underline{2} = 0$$

$$\Rightarrow \underline{x^4 + 1 - 3x^3 - 3x - 2x^2} = 0$$

$$D = 4$$

$$0 \in \mathbb{R}$$

$$(vii) x + 1/x = x^2, \quad x \neq 0$$

$$\Rightarrow \frac{x}{1} + \frac{1}{x} = x^2$$

$$\Rightarrow \frac{x^2 + 1}{x} = x^2$$

$$\Rightarrow x^2 + 1 = x^2 \cdot x$$

$$\Rightarrow x^2 + 1 = x^3$$

$$\Rightarrow \boxed{-x^3 + x^2 + 1 = 0}$$

$$D = 3$$

$$Q \in \mathbb{K}$$

$$(viii) 16x^2 - 3 = (2x + 5)(5x - 3)$$

$$\Rightarrow 16x^2 - 3 = \underbrace{10x^2}_{+10x} - \underbrace{6x + 25x}_{+19x} - 15$$

$$\Rightarrow 16x^2 - 10x^2 - 3 + 15 - 19x = 0$$

$$\Rightarrow \boxed{6x^2 - 19x + 12 = 0}$$

$$D = 2$$

$$Q \in \mathbb{K} \checkmark$$

#Solutions of a Quadratic Equation:

zeros / roots

→ variable की वो value जिससे $Q.E = 0$ हो जाए

Eg:- $x^2 - 4$, find solution

$$x = 2$$

$$\Rightarrow (2)^2 - 4$$

$$= 4 - 4$$

$$= 0$$

$$\Rightarrow x^2 - 4 = 0$$

$$\Rightarrow (x)^2 - (2)^2 = 0$$

$$(x+2)(x-2) = 0$$

$$x+2=0$$

$$x = -2$$

$$x-2=0$$

$$x = 2$$

$$x^2 = 4$$

$$x = \sqrt{4}$$

$$x = \pm 2$$

#LP: In each of the following, determine whether the given values are solutions of the given equation or not:

(i) $x^2 - 3x + 2 = 0$, $x = 2$, $x = -1$

checking $x = 2$

$$\Rightarrow (2)^2 - 3(2) + 2$$

$$\Rightarrow 4 - 6 + 2$$

$$\Rightarrow -2 + 2$$

$$\Rightarrow 0$$

$\therefore x = 2$ is a soln.

$$x = -1$$

$$\Rightarrow (-1)^2 - 3(-1) + 2$$

$$\Rightarrow 1 + 3 + 2$$

$$\Rightarrow 6 \neq 0$$

$x = -1$ is not a soln.

(ii) $\underline{x^2} - 3\sqrt{3}\underline{x} + 6 = 0$, $x = \sqrt{3}$ and $x = -2\sqrt{3}$

check $x = \sqrt{3}$

$$\Rightarrow (\sqrt{3})^2 - 3\sqrt{3}(\sqrt{3}) + 6$$

$$\Rightarrow 3 - 3 \times 3 + 6$$

$$= 3 - 9 + 6$$

$$= -6 + 6$$

$$= 0$$

$x = \sqrt{3}$ is a sol ✓

check $x = -2\sqrt{3}$

$$\Rightarrow (-2\sqrt{3})^2 - 3\sqrt{3}(-2\sqrt{3}) + 6$$

$$\Rightarrow (-2)^2 (\sqrt{3})^2 - 3 \cdot \sqrt{3} \times -2\sqrt{3} + 6$$

$$\Rightarrow 4 \times 3 + 6 \times 3 + 6$$

$$\Rightarrow 12 + 18 + 6$$

$$\Rightarrow 36 \neq 0$$

$x = -2\sqrt{3}$ is not a sol ✗

$$(ab)^n = a^n b^n$$

$$(a+b)^2 \neq a^2 + b^2$$

(iii) $a^2 x^2 - 3abx + 2b^2 = 0$, $x = a/b$, $x = b/a$ ✓

$x = \frac{a}{b}$

$$\Rightarrow a^2 \left(\frac{a}{b}\right)^2 - 3ab \left(\frac{a}{b}\right) + 2b^2$$

$$\Rightarrow a^2 \times \frac{a^2}{b^2} - 3a^2 + 2b^2$$

$$\Rightarrow \frac{a^4}{b^2} - 3a^2 + 2b^2$$

$\neq 0$ $x = \frac{a}{b}$ is not sol.

$$x = \frac{b}{a} \Rightarrow a^2 \left(\frac{b}{a}\right)^2 - 3ab \left(\frac{b}{a}\right) + 2b^2$$

$$\Rightarrow \cancel{a^2} \times \frac{b^2}{\cancel{a^2}} - 3b^2 + 2b^2$$

$$\Rightarrow \cancel{b^2} - \cancel{b^2}$$

$$\Rightarrow 0$$

$\therefore x = \frac{b}{a}$ is a sol ✓

~~hw~~
(iv) $x + 1/x = 13/6$, $x = 5/6$, $x = 4/3$

HW → Module
PL 2V

Thank You

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